Quasi-local diffeomorphism charges in a closed S^3 universe: explicit Iyer–Wald charges, the Gibbons–Hawking–York term, and ϕ_{g}-level boundary scanning

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Diffeomorphism invariance implies Noether identities and first-class constraints rather than ordinary global conserved charges. In a compact closed universe with Cauchy slice $\Sigma \simeq S^3$ one has $\partial\Sigma = \varnothing$, hence the global Hamiltonian generator associated with gauge diffeomorphisms vanishes. Nevertheless, physically meaningful generators arise quasi-locally for subregions $D \subset \Sigma$ with $\partial D \neq \varnothing$ and are entirely encoded by boundary integrals. We provide a self-contained covariant phase space derivation in general relativity with an explicit Iyer–Wald Noether charge two-form Q_ξ and an explicit boundary three-form B corresponding to the Gibbons–Hawking–York term. For closed FRW geometry we compute the Brown–York energy of geodesic balls in $\Sigma \simeq S^3$ and obtain a closed expression in terms of the areal radius R and curvature radius $a(t)$. We relate this boundary energy to the FRW Misner–Sharp mass and clarify that the two coincide only after additional dynamical input (e.g. Friedmann constraints and/or a choice of evolution vector field), since Brown–York is tied to a chosen slice and evolution vector field, whereas Misner–Sharp is the invariant quasi-local mass in spherical symmetry and is often associated with the Kodama flow.[15, 16] We also highlight constant-mean-curvature slicing and York time as a canonical choice of relational clock in compact universes. Finally, we introduce a ϕ_{g}-discretized boundary scan (a golden-ratio discretization of radii, not a dynamical scalar field) $R_n = R_0\phi_{\mathrm{g}}^n$ as a structured quasi-local bookkeeping device; a model-dependent $\zeta_{\phi_{\mathrm{g}}}$ spectral overlay and a modular reflection map $n \leftrightarrow n'$ are recorded separately in an appendix.

I. INTRODUCTION

In diffeomorphism-invariant theories, conservation laws take a different structural form from those of global symmetries. Noether's second theorem implies identities among the Euler–Lagrange equations and yields first-class constraints. In general relativity (GR), the full compact-universe logic may be summarized in the one-line chain

$$\boxed{\begin{array}{c}\mathrm{Diff}(\mathcal{M}) \Rightarrow \text{Noether II} \Rightarrow \boldsymbol{J}_\xi \approx \mathrm{d}\boldsymbol{Q}_\xi \\ H_\xi = \displaystyle\int_{\partial\Sigma} \left(\boldsymbol{Q}_\xi - \xi\cdot\boldsymbol{B}\right) \Rightarrow \partial\Sigma = \varnothing \\ \Rightarrow H_\xi[\Sigma] = 0(\text{gauge } \xi)\end{array}} \tag{1}$$

where $\approx$ denotes equality on-shell. The key point for cosmology is that a spatially closed universe with compact Cauchy slice $\Sigma \simeq S^3$ has no boundary. Hence there is no nontrivial *global* boundary generator for *gauge* diffeomorphisms. Physical information is encoded (i) in constraints, and (ii) in quasi-local boundary charges for subregions $D \subset \Sigma$. This paper makes these statements explicit in the covariant phase space formalism and computes the resulting quasi-local energy in closed FRW geometry. Throughout, ξ^a denotes a smooth vector field used as an *evolution/gauge parameter* in the covariant phase space formalism. Unless ξ is a genuine symmetry (e.g. Killing), the associated quasi-local generator need not be conserved under time evolution; the construction yields a meaningful Hamiltonian generator on a chosen phase space with specified boundary conditions, not automatically a globally conserved charge in the Noether-I sense.

II. FROM NOETHER I TO NOETHER II, BOUNDARY CHARGES, AND ADM

It is useful to situate the chain (1) within the more familiar Noether-I picture. For ordinary global symmetries (finite-dimensional parameters), Noether's first theorem yields a conserved current $\nabla_\mu j^\mu = 0$ and an associated charge obtained by integrating j^μ over a Cauchy slice. In that setting the charge is genuinely "bulk": its conservation follows from the local continuity equation together with Stokes' theorem. Diffeomorphisms, by contrast, are a local gauge redundancy with an arbitrary spacetime-dependent parameter $\xi^\mu(x)$. Noether's second theorem therefore applies: invariance under $\xi^\mu(x)$ produces identities among the Euler–Lagrange equations (Bianchi-type relations) and, in the Hamiltonian language, first-class constraints. In covariant phase space (CPS) this shows up as a strengthening of "current conservation": the diffeomorphism Noether current is on-shell exact,

$$\boldsymbol{J}_\xi \approx \mathrm{d}\boldsymbol{Q}_\xi, \tag{2}$$

so the would-be charge is necessarily encoded by a surface integral. More precisely, writing the Lagrangian as a top form $\boldsymbol{L}$ and its variation as

$$\delta\boldsymbol{L} = E(\phi)\,\delta\phi + \mathrm{d}\boldsymbol{\theta}(\phi;\delta\phi), \tag{3}$$

one defines

$$\boldsymbol{J}_\xi := \boldsymbol{\theta}(\phi;\mathcal{L}_\xi\phi) - \xi\cdot\boldsymbol{L}. \tag{4}$$

The CPS symplectic current $\boldsymbol{\omega} = \delta\boldsymbol{\theta}$ then gives the variation of the Hamiltonian generator as a boundary term,

$$\delta H_\xi[D] = \int_{\partial D} \Big(\delta\boldsymbol{Q}_\xi - \xi\cdot\boldsymbol{\theta}(\phi;\delta\phi)\Big), \tag{5}$$

for any spatial region $D \subset \Sigma$. If the boundary conditions and the choice of phase space imply integrability,

$$\xi\cdot\boldsymbol{\theta}(\phi;\delta\phi) = \delta(\xi\cdot\boldsymbol{B}) \quad \text{on the chosen phase space,} \tag{6}$$

then one obtains an actual generator

$$H_\xi[D] = \int_{\partial D} \big(\boldsymbol{Q}_\xi - \xi\cdot\boldsymbol{B}\big), \tag{7}$$

up to an additive constant. In general, non-integrability can occur (physically: boundary flux); addressing it may require additional structure (e.g. Wald–Zoupas-type prescriptions[12]). In this work we restrict attention to boundary data for which (6) holds. Concretely, we assume Dirichlet boundary conditions for the induced metric on the timelike worldtube swept out by ∂D, so that the GHY choice makes the Hamiltonian integrable on the chosen phase space. The ADM formulation makes the same statement in canonical language. Decomposing $\xi^\mu = Nn^\mu + N^\mu$ into lapse N and shift N^i, the canonical generator takes the form

$$H_{\rm ADM}[N,N^i] = \int_\Sigma \big(N\,\mathcal{H} + N^i\mathcal{H}_i\big) \;+\; \oint_{\partial\Sigma} \mathcal{B}[N,N^i]. \tag{8}$$

Here $\mathcal{H} \approx 0$ and $\mathcal{H}_i \approx 0$ are the first-class Hamiltonian and momentum constraints implied by Noether II, while $\mathcal{B}$ is the boundary functional needed for differentiability and for nonzero charges when a boundary exists. On-shell the bulk term vanishes and the generator is purely boundary-supported. Therefore, in a closed universe with compact $\Sigma \simeq S^3$ (or more generally any compact Σ without boundary),

$$\partial\Sigma = \varnothing \qquad\Longrightarrow\qquad H_\xi[\Sigma] = 0 \tag{9}$$

for all smooth *gauge* diffeomorphisms, while quasi-local charges can still be defined for subregions $D \subset \Sigma$ with $\partial D \neq \varnothing$.

A. Closed FRW in one line: $\partial\Sigma = \varnothing \Rightarrow H_\xi[\Sigma] = 0$ (gauge ξ)

In a closed FRW cosmology $\mathcal{M} = \mathbb{R}\times S^3$ one may write

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a(t)^2\,\mathrm{d}\Omega_3^2, \qquad \Sigma_t \simeq S^3, \tag{10}$$

so every Cauchy slice is compact and boundaryless: $\partial\Sigma_t = \varnothing$. Therefore

$$H_\xi[\Sigma_t] = \int_{\partial\Sigma_t} \big(\boldsymbol{Q}_\xi - \xi\cdot\boldsymbol{B}\big) = 0, \tag{11}$$

for any smooth diffeomorphism parameter ξ interpreted as a gauge direction on the phase space.

B. Explicit Q_ξ (Komar/Iyer–Wald) and a quasi-local $S^2 \subset S^3$ example

For Einstein–Hilbert gravity, the Iyer–Wald Noether charge two-form is

$$(\boldsymbol{Q}_\xi)_{ab} = -\frac{1}{16\pi G}\,\epsilon_{abcd}\,\nabla^c\xi^d. \tag{12}$$

Even though the global slice has no boundary, quasi-local charges are nontrivial for subregions. Take a geodesic ball $D \subset \Sigma_t \simeq S^3$ with boundary $\partial D \simeq S^2$ (e.g. $\chi = \chi_0$ in hyperspherical coordinates), and choose $\xi = u = \partial_t$ (the comoving time flow). Then the quasi-local generator is a boundary integral on ∂D,

$$H_u[D] = \int_{\partial D} \big(\boldsymbol{Q}_u - u\cdot\boldsymbol{B}\big). \tag{13}$$

With a suitable boundary term $\boldsymbol{B}$ and a reference subtraction fixing the zero-point, this reproduces the Brown–York energy computed below. Since u is not generally Killing in FRW, $H_u[D]$ should be viewed as a quasi-local Hamiltonian associated with the chosen slicing/evolution, not as an a priori conserved charge.

III. COVARIANT PHASE SPACE PRELIMINARIES

We follow the covariant phase space formalism of Lee–Wald and the diffeomorphism-charge construction of Iyer–Wald.[3, 11] Let $\boldsymbol{L}(\phi)$ be a Lagrangian 4-form on spacetime $\mathcal{M}$. Its variation is

$$\delta\boldsymbol{L} = E(\phi)\,\delta\phi + \mathrm{d}\boldsymbol{\theta}(\phi;\delta\phi), \tag{14}$$

where $E(\phi) = 0$ are equations of motion and $\boldsymbol{\theta}$ is the symplectic potential 3-form. The symplectic current 3-form is

$$\boldsymbol{\omega}(\phi;\delta_1\phi,\delta_2\phi) = \delta_1\boldsymbol{\theta}(\phi;\delta_2\phi) - \delta_2\boldsymbol{\theta}(\phi;\delta_1\phi). \tag{15}$$

For a vector field ξ^a, diffeomorphism invariance implies $\delta_\xi\boldsymbol{L} = \mathcal{L}_\xi\boldsymbol{L} = \mathrm{d}(\xi\cdot\boldsymbol{L})$.

IV. NOETHER CURRENT, NOETHER II, AND BOUNDARY HAMILTONIANS

Define the Noether current 3-form

$$\boldsymbol{J}_\xi := \boldsymbol{\theta}(\phi;\mathcal{L}_\xi\phi) - \xi\cdot\boldsymbol{L}. \tag{16}$$

A standard identity yields $\mathrm{d}\boldsymbol{J}_\xi = -E(\phi)\mathcal{L}_\xi\phi$, so $\mathrm{d}\boldsymbol{J}_\xi \approx 0$. For gauge symmetries such as diffeomorphisms, Noether II strengthens this to on-shell exactness:

$$\boldsymbol{J}_\xi \approx \mathrm{d}\boldsymbol{Q}_\xi, \tag{17}$$

for some locally constructed 2-form $\boldsymbol{Q}_\xi$. In the covariant phase space formalism, the variation of the Hamiltonian

generator associated with ξ over a spatial region $D \subset \Sigma$ reduces to a boundary expression (5). When (6) holds for the chosen boundary conditions, the integrable generator is given by (7).

V. EXPLICIT GR FORMULAS: IYER–WALD Q_ξ AND THE GHY BOUNDARY FORM B

We now specialize to GR (with optional Λ). Let $\boldsymbol{\epsilon}$ denote the spacetime volume 4-form. The Einstein–Hilbert Lagrangian is

$$\boldsymbol{L}_{\rm EH} = \frac{1}{16\pi G}(R - 2\Lambda)\,\boldsymbol{\epsilon}. \tag{18}$$

In Iyer–Wald conventions, the symplectic potential may be written as

$$\boldsymbol{\theta}_{abc}(g;\delta g) = \frac{1}{16\pi G}\,\epsilon_{dabc}\Big(\nabla_e \delta g^{de} - \nabla^d \delta g\Big), \tag{19}$$

where $\delta g := g_{mn}\delta g^{mn}$ and indices are moved with g_{ab}. The Iyer–Wald Noether charge 2-form is (12). To make the variational principle well-posed on manifolds with a spacetime boundary hypersurface $\mathcal{B}$ under Dirichlet boundary conditions (fixed induced metric on $\mathcal{B}$), one adds the Gibbons–Hawking–York term,[13, 14]

$$S_{\rm GHY} = \frac{1}{8\pi G}\int_{\mathcal{B}} \sqrt{|h|}\,K\,\mathrm{d}^3x, \tag{20}$$

corresponding to the boundary 3-form

$$\boldsymbol{B}_{\rm GHY} = \frac{1}{8\pi G}\,K\,\boldsymbol{\epsilon}_{\mathcal{B}}. \tag{21}$$

Here K is the trace of the extrinsic curvature of the *3D* boundary $\mathcal{B} \subset \mathcal{M}$ in spacetime, and $\boldsymbol{\epsilon}_{\mathcal{B}}$ is its induced volume form. In quasi-local Hamiltonians one ultimately evaluates boundary functionals on *2-surfaces* $\partial D \subset \Sigma$; a reference subtraction (or counterterm) is additionally required to fix the energy zero-point, as in the Brown–York construction below.

VI. CLOSED FRW GEOMETRY AND QUASI-LOCAL ENERGY ON $\Sigma \simeq S^3$

Consider a closed FRW slicing with spatial metric on $\Sigma_t \simeq S^3$:

$$\mathrm{d}\ell^2 = a(t)^2\left(\mathrm{d}\chi^2 + \sin^2\chi\,\mathrm{d}\Omega^2\right), \qquad \chi \in [0,\pi]. \tag{22}$$

Let $D(\chi_0)$ be a geodesic ball $0 \le \chi \le \chi_0$ with boundary $\partial D \simeq S^2$ at $\chi = \chi_0$. The areal radius and area are

$$R(\chi_0) = a(t)\sin\chi_0, \qquad A(\chi_0) = 4\pi R^2, \tag{23}$$

and the volume is

$$V(\chi_0) = 2\pi a(t)^3\big(\chi_0 - \sin\chi_0\cos\chi_0\big). \tag{24}$$

The Brown–York energy of D is

$$E_{\rm BY} = \frac{1}{8\pi G}\int_{\partial D}(k_0 - k)\,\mathrm{d}S, \tag{25}$$

where k is the trace of the extrinsic curvature of ∂D embedded in Σ (with outward unit normal in Σ) and k_0 is a reference term fixing the zero-point. A standard choice is the Euclidean reference: k_0 is taken as the mean curvature of an isometric embedding of the same intrinsic 2-metric into $\mathbb{R}^3$, giving $k_0 = 2/R$. For the round S^2 boundary with intrinsic metric $R^2 d\Omega^2$, the standard isometric embedding into $\mathbb{R}^3$ exists and yields this value. For $\chi = \chi_0$ spheres in $\Sigma \simeq S^3$,

$$k(\chi_0) = \frac{2}{a}\cot\chi_0 = \frac{2}{R}\sqrt{1 - \frac{R^2}{a^2}}, \qquad k_0 = \frac{2}{R}. \tag{26}$$

Substituting into (25) yields the closed expression

$$E_{\rm BY}(R,a) = \frac{R}{G}\left(1 - \sqrt{1 - \frac{R^2}{a^2}}\right), \qquad 0 \le R \le a(t). \tag{27}$$

This result was previously obtained in the equivalent coordinate system by Afshar [8]. At the equator ($R \to a$) one has $E_{\rm BY} \to R/G$. For small balls $R \ll a$,

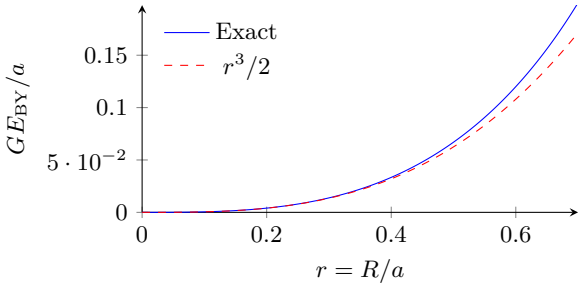


FIG. 1. Brown–York quasi-local energy in closed FRW cosmology: $GE_{\rm BY}/a$ versus $r = R/a$ (solid curve). The small-ball approximation $r^3/2$ is shown dashed for comparison.

$$E_{\rm BY}(R,a) = \frac{R^3}{2Ga^2} + O\!\left(\frac{R^5}{a^4}\right). \tag{28}$$

This small-ball limit highlights that, even before invoking dynamics, the positive spatial curvature of S^3 induces a nontrivial quasi-local energy relative to the Euclidean reference.

VII. MISNER–SHARP MASS, FRIEDMANN CONSTRAINT, AND COMPARISON WITH BROWN–YORK

In spherically symmetric spacetimes, the Misner–Sharp mass provides an invariant quasi-local energy: It is often

associated with the Kodama vector field, which supplies a preferred energy current in spherical symmetry.

$$M_{\rm MS}(R) = \frac{R}{2G}\left(1 - g^{ab}\nabla_a R\,\nabla_b R\right). \tag{29}$$

For FRW, this becomes

$$M_{\rm MS}(R) = \frac{R^3}{2G}\left(H^2 + \frac{k}{a^2}\right), \qquad H := \frac{\dot{a}}{a}. \tag{30}$$

For the closed case ($k = +1$),

$$M_{\rm MS}(R) = \frac{R^3}{2G}\left(H^2 + \frac{1}{a^2}\right). \tag{31}$$

The Friedmann constraint reads

$$H^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \tag{32}$$

hence

$$M_{\rm MS}(R) = \frac{4\pi}{3}R^3\rho + \frac{\Lambda R^3}{6G}. \tag{33}$$

Equation (27) depends only on the intrinsic/extrinsic geometry of the 2-surface ∂D as embedded in the chosen spatial slice Σ_t (plus the chosen reference subtraction), while (30) is a spacetime-covariant quantity depending on $\nabla_a R$ and thus on expansion via H. Accordingly, $E_{\rm BY}$ and $M_{\rm MS}$ should be viewed as complementary quasi-local energies: $E_{\rm BY}$ is naturally tied to a slicing and boundary conditions, whereas $M_{\rm MS}$ is invariant in spherical symmetry and directly relates to matter content through (33). A precise identification between them requires additional dynamical input and/or a preferred evolution vector field, and is not automatic at the level of purely kinematical definitions.

VIII. CMC SLICING, YORK TIME, AND RELATIONAL OBSERVABLES IN A COMPACT UNIVERSE

In a compact universe, $H_\xi[\Sigma] = 0$ expresses that the total Hamiltonian is a sum of first-class constraints and vanishes on-shell. To interpret dynamics, one introduces a relational clock. For general discussions of relational clocks and relational observables in Hamiltonian constrained systems, see e.g. [9, 10]. A widely used choice is constant-mean-curvature (CMC) slicing. Let h_{ij} be the induced metric on Σ_t and K_{ij} its extrinsic curvature in spacetime; define the mean curvature

$$K := h^{ij}K_{ij}. \tag{34}$$

CMC slicing imposes $K = \text{const}$ on each Σ_t. For standard FRW slicings, K is spatially constant automatically by homogeneity and isotropy, so the usual cosmological foliation is a CMC foliation. York time is then defined (up to conventions) by

$$T_Y := -\frac{1}{3}K. \tag{35}$$

In compact universes, such a choice supplies a canonical relational time variable compatible with constraint dynamics, providing a clean way to parametrize evolution despite the vanishing of the global Hamiltonian generator.

IX. $\phi_{\rm g}$-LEVEL BOUNDARY SCANNING AND A ONE-BLOCK CORRESPONDENCE MAP

Define a $\phi_{\rm g}$-discretized family of areal radii (where $\phi_{\rm g} := (1+\sqrt{5})/2$ is the golden ratio)

$$\begin{aligned} R_n &= R_0\phi_{\rm g}^n, \\ n &= \frac{\log(R/R_0)}{\log\phi_{\rm g}}. \end{aligned} \tag{36}$$

restricted in $\Sigma \simeq S^3$ by $0 \le R_n \le a(t)$. This discretization is introduced purely as a structured quasi-local bookkeeping device and does not represent a claim that GR dynamically selects the constant $\phi_{\rm g}$. Then

$$\begin{aligned} E_{\rm BY}(n) &= \frac{R_n}{G}\left(1 - \sqrt{1 - \frac{R_n^2}{a^2}}\right), \\ M_{\rm MS}(n) &= \frac{R_n^3}{2G}\left(H^2 + \frac{1}{a^2}\right). \end{aligned} \tag{37}$$

X. DISCRETE S^3 EVOLUTION AND GOLDEN-RATIO CONSISTENCY (SUMMARY OF THE FBS^{3}R CORE)

We briefly summarize the structural assumptions of the discrete evolution model developed in a companion discrete-framework note, restricting to the minimal elements needed to interface with the quasi-local diffeomorphism-charge analysis of Secs. II–VI. The purpose of this section is *not* to re-derive the FBS^{3}R construction, but only to record its postulates and to make the compatibility with covariant phase space (CPS) and GR explicit.

A. Postulates (minimal)

a. (P1) Topological ontology. At each discrete stage the spatial manifold is compact, simply connected, and boundaryless:

$$\Sigma_n \simeq S^3, \qquad \partial\Sigma_n = \varnothing, \qquad \Sigma_n \cong \Sigma_{n+1}. \tag{38}$$

No topology change is assumed.

b. (P2) Causal continuity (internal evolution). Evolution is internal (no external embedding space is required) and is represented as a discrete update

$$\text{state}_n \to \text{state}_{n+1}, \tag{39}$$

with the spatial topology fixed as in (P1).

c. (P3) Fixed total energy. A conserved total energy is postulated,

$$E_{\text{total}} = \text{const}, \qquad \rho_n = \frac{E_{\text{total}}}{V_n}, \tag{40}$$

where V_n is the model-assigned volume associated with the radius parameter introduced below. (When comparing to GR expressions on $\Sigma \simeq S^3$, one may instead use the geometric S^3 volume of a geodesic ball, cf. Sec. VII; the present subsection merely records the postulated bookkeeping used in a companion discrete-framework note.)

B. Discrete geometric scaling

The discrete radius is taken to scale as

$$R_n = \ell_P\, \phi_{\text{g}}^{N_n}, \qquad \phi_{\text{g}} := \frac{1+\sqrt{5}}{2}, \tag{41}$$

with integer level N_n (possibly time-dependent through the discrete evolution). A corresponding volume scaling is assigned in the FBS^{3}R model as $V_n \propto R_n^3$, hence

$$\rho_n \propto \phi_{\text{g}}^{-3N_n}. \tag{42}$$

This provides a compact encoding of density scaling compatible with the fixed-energy postulate (P3).

C. Quantum surface growth and Fibonacci asymptotics

The number of "surface quanta" obeys a Fibonacci recursion,

$$N_{n+1} = N_n + N_{n-1}, \tag{43}$$

so that asymptotically

$$\lim_{n\to\infty} \frac{N_{n+1}}{N_n} = \phi_{\text{g}}. \tag{44}$$

In particular, this yields an asymptotic ϕ_{g}-ratio between successive radii,

$$\frac{R_{n+1}}{R_n} = \phi_{\text{g}}^{N_{n+1}-N_n} \quad \xrightarrow[n\to\infty]{} \quad \phi_{\text{g}}, \tag{45}$$

establishing golden-ratio consistency between (i) discrete geometric scaling, (ii) Fibonacci growth, and (iii) surface-based bookkeeping of density levels. (a companion discrete-framework note also introduces the normalization $F_n = \phi_{\text{g}}^n/\sqrt{5}$; we do not need it for the GR/CPS derivations in the present paper.)

D. Compatibility with diffeomorphism invariance and CPS/GR charges

Because each Σ_n is diffeomorphic to S^3 and is globally boundaryless, the Hamiltonian generators of *gauge* diffeomorphisms vanish on the full slice:

$$H_\xi[\Sigma_n] = \int_{\partial\Sigma_n} \big(\boldsymbol{Q}_\xi - \xi\cdot\boldsymbol{B}\big) = 0. \tag{46}$$

This is precisely the compact-universe statement emphasized in Eq. (1). Nevertheless, for subregions $D \subset \Sigma_n$ with $\partial D \neq \varnothing$, quasi-local Hamiltonians remain well-defined and are encoded by boundary integrals,

$$H_\xi[D] = \int_{\partial D} \big(\boldsymbol{Q}_\xi - \xi\cdot\boldsymbol{B}\big), \tag{47}$$

with $(\boldsymbol{Q}_\xi, \boldsymbol{B})$ as constructed explicitly in GR in Secs. V–VI. Thus, the discrete ϕ_{g}-driven scaling recorded in (41)–(45) does not conflict with diffeomorphism invariance: it specifies a relational/discrete parametrization of evolution within a fixed diffeomorphism class $\Sigma \simeq S^3$, while the CPS/GR charge algebra and the statement $H_\xi[\Sigma] = 0$ remain intact.

E. York time as a relational parameter

In compact $\Sigma \simeq S^3$ settings, dynamics is naturally described in relational terms. A standard canonical choice is constant-mean-curvature (CMC) slicing and York time,

$$T_Y := -\frac{1}{3}K, \tag{48}$$

with K the mean curvature of the embedding of Σ in spacetime (Sec. IX). Within this viewpoint, the discrete index n (or level N_n) may be treated as a monotonic function of T_Y,

$$n = n(T_Y), \tag{49}$$

providing a relational clock compatible with the vanishing of global gauge generators on a closed slice.

XI. CONCLUSIONS

In a compact closed universe $\Sigma \simeq S^3$, diffeomorphism invariance implies that global gauge diffeomorphism generators vanish: $H_\xi[\Sigma] = 0$. Nontrivial physics is carried by constraints and by quasi-local boundary charges for subregions. We made this explicit in GR by providing the closed-form Iyer–Wald charge $\boldsymbol{Q}_\xi$ and the GHY boundary form $\boldsymbol{B}$, together with the standard integrability condition needed to define a Hamiltonian generator. For closed FRW geometry, we computed the Brown–York energy of geodesic balls and related it to the Misner–Sharp mass and

the Friedmann constraint, emphasizing the complementary (slice-adapted vs spacetime-covariant) character of these notions. CMC slicing and York time supply a canonical relational clock compatible with compact-universe constraint dynamics. Finally, a $\phi_{\rm g}$-discretized boundary scan offers a structured quasi-local bookkeeping scheme; model-dependent spectral overlays can be layered without altering the GR derivations.

Appendix A: Derivation notes and ambiguities

The covariant phase space formalism includes standard ambiguities: $\boldsymbol{L} \to \boldsymbol{L} + \mathrm{d}\boldsymbol{\mu}$ and $\boldsymbol{\theta} \to \boldsymbol{\theta} + \delta\boldsymbol{\mu} + \mathrm{d}\boldsymbol{\nu}$. These induce shifts in $(\boldsymbol{J}_\xi, \boldsymbol{Q}_\xi)$ and in the definition of the Hamiltonian by boundary terms. The choice of $\boldsymbol{B}$ (including the GHY term and a reference subtraction/counterterm) fixes both integrability and the zero-point of quasi-local energy.

Appendix B: Extrinsic curvature on S^3

For $\chi = \text{const}$ spheres, the unit normal within Σ is

$$n_i = \frac{1}{a}\,(\partial/\partial\chi)_i. \tag{B1}$$

A direct computation yields

$$k = \sigma^{AB}\nabla_A n_B = \frac{2}{a}\cot\chi, \tag{B2}$$

where σ_{AB} is the induced metric on the 2-sphere ∂D and ∇ is the Levi–Civita connection of Σ.

Appendix C: $\zeta_{\phi_{\rm g}}$ spectral overlay and modular reflection map (model-dependent)

This appendix records a phenomenological overlay used in $\phi_{\rm g}$-hierarchical modeling; it does not modify any GR statements in the main text and is logically independent of the quasi-local derivations. Define an energy ladder

$$E_n^{(\zeta)} := \frac{\hbar c}{\ell_p\,\phi_{\rm g}^n}, \tag{C1}$$

with corresponding $m_n^{(\zeta)} = E_n^{(\zeta)}/c^2$. Fix a distinguished index N and define a reflection map $n' := N - n$.

Appendix D: Explicit Killing generators on S^3 (embedding form)

Let S^3 be the unit sphere embedded in $\mathbb{R}^4$ with Cartesian coordinates X^A obeying $X^A X^A = 1$. The full isometry group is SO(4), generated by six Killing fields J_{AB}:

$$J_{AB} := X^A\frac{\partial}{\partial X^B} - X^B\frac{\partial}{\partial X^A}, \qquad 1 \le A < B \le 4. \tag{D1}$$

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